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SUPERSEDING
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7 AUGUST 1956

MILITARY SPECIFICATION

POWER METER
(POWER METER TS-125( )/AP)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a compact, battery-operated, portable radar test set designated as Power Meter TS-125( )/AP (see 6.4).

1.2 The Power Meter consists of the following units:

Basic Unit for Power Meter TS-125 ( )/AP.
Attenuator CN-42( )/UP.
Attenuator CN-43( )/UP.
Cord CG-299A/U (-ft.-in.).
Antenna AT-67( )/AP.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-P-515 — Boxes, Wood, Nailed and Lock-Corner.
PPP-B-291 — Plywood, Container Grade.
PPP-B-585 — Paperboard, Wrapping Cushioning.
PPP-B-591 — Boxes, Wood, Wirebound.
PPP-B-601 — Boxes, Fiberboard, Wood-Cleated.
PPP-B-621 — Boxes, Wood, Cleated Plywood.
PPP-B-636 — Boxes, Fiber.
PPP-T-60 — Tape, Pressure Sensitive Adhesive Water Proof for Packaging and Sealing.

MILITARY

MIL-M-6 — Meters, Electrical Indicating, Panel Type (2½ in. and 3½ in.), General Specification for.
MIL-B-18 — Batteries, Dry.
MIL-P-116 — Preservation, Methods of.

FSC 6625

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MIL-B-117 — Bags, Interior, Packaging.

MIL-T-152 — Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.

MIL-T-945 — Test Equipment for use with Electronic Equipment; General Specification for.

MIL-A-4637 — Attenuator CN-29 ( )/UP, CN-42 ( )/UP, CN-43 ( )/UP.

MIL-M-10304 — Meters, Electrical Indicating Panel Type, Ruggedized.

MIL-M-13231 — Marking of Electronic Items.

MIL-F-14072 — Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-108 — Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-252 — Wired Equipment, Classification of Visual and Mechanical Defects for.

MS91231 — Connector, Plug, Electrical, Series N, UG-18C/U.

DRAWINGS

SIGNAL CORPS

SC-DL-20936 — Cord CG-299A/U (-ft. -in.).

SC-DL-166805-A — Power Meter, Test Set TS-125/AP.

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification.

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth St., N. W., Washington 6, D. C.)

U. S. POST OFFICE DEPARTMENT

Parcel Post Regulations.

(Application for copies should be obtained from any 1st class U. S. Post Office.)

3. REQUIREMENTS

3.1 Description. Power Meter TS-125( )/AP is in a cast aluminum case and consists of a microammeter connected in a thermistor bridge circuit, and three standard flashlight batteries. A pickup horn antenna and an RF cable are stored in the lid of the instrument, and two attenuators are stored within the case. The meter gives full-scale deflection when 2 milliwatts of RF power are applied at the input connector. Two scales are provided; one reads milliwatts (mw) of power and the other, decibels above and below one milliwatt (dbm). An ON-OFF switch and balancing potentiometer are operated by a single control located on the front of the instrument below the meter.

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3.2 Construction. Power Meter TS-125 ( )/AP shall be constructed in accordance with Drawing SC-DL-166805 and as specified herein. In case of conflict between the requirements of this specification and Drawing SC-DL-166805, this specification shall govern.

3.2.1 *Power meter.* Construction of the power meter for Power Meter TS-125( )/AP shall be in accordance with Drawing SC-D-166806, covering a detail of SC-DL-166805.

3.2.2 *Attenuators CN-42( )/UP and CN-43( )/UP.* Attenuators CN-42( )/UP and CN-43( )/UP shall be constructed in accordance with Specification MIL-A-4637.

3.2.3 *Cord CG-299A/U (-ft. -in.).* Cord CG-299A/U (-ft. in.) shall consist of a 9-foot 3-inch length of Cable, Radio Frequency RG-21A/U constructed in accordance with drawing SC-DL-20936. Each cable shall be marked indicating attenuation and calibration.

3.2.4 *Antenna AT-67( )/AP.* Construction of Antenna AT-67( )/AP shall be in accordance with drawing SC-C-45551 covering a detail of drawing SC-DL-166805. The antenna shall be capable of withstanding immersion and shall be stored in the lid of the case for Power Meter TS-125( )/AP when not in use.

3.3 Parts, materials, and processes; general. In addition to any requirements of this specification, covering parts, materials, and processes, such items shall conform to Specification MIL-T-945 including the selection requirements therein, as applicable. In case of conflict between the requirements of this specification and Specification MIL-T-945, this specification shall govern.

3.3.1 *Spare parts.* A spare thermistor, Western Electric Part Number 163903 or equal, and a spare meter shunt resistor, Electron Radar Part drawing No. MEPCO M-52 or equal, shall be supplied and secured inside the case. The resistor and thermistor shall be a selected pair which will give 2 mw full-scale meter reading.

3.3.2 *Batteries.* The Power Meter TS-125 ( )/AP shall function satisfactorily with an input voltage of 4.5 volts dc as provided by three BA-30 type batteries in accordance with Specification MIL-B-18. The contractor shall not be required to furnish batteries and no equipment shall be supplied with batteries installed.

3.3.3 *Battery housing.* Provision shall be made for easy removal of swollen batteries.

3.3.4 *Battery housing cap.* The battery housing cap shall be watertight.

3.3.5 *Meter.* The meter shall meet the requirements of Specification MIL-M-6, and the shock and vibration specified in Specification MIL-M-10304 for ruggedized meters.

3.3.6 *Tropicalization.*

3.3.6.1 *Treatment of assembled equipment.* Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc.) which is applicable on contract.

(2) It does not incorporate any material listed in 3.3.6.2 unless the material has been given the treatment specified in Specification MIL-T-945 or a treatment approved by the contracting officer.

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(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL–F–14072.

3.3.6.2 *Material not resistant to moisture and fungi.* The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.

(b) Cotton.

(c) Cork.

(d) Felt, hair or wool.

(e) Jute.

(f) Leather.

(g) Linen.

(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.

(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.3.6.3 *Statement of treatment.* The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. This statement shall include the statements required by Specification MIL–T–945 and MIL–T–152, as applicable.

3.4 Electrical requirements.

3.4.1 *Power range.* The equipment shall be capable of measuring average power over a range of ±10 dbm to +26 dbm. The upper limit includes the 16-db attenuation of Attenuator CN–43( )/UP and the 7-db attenuation of the Cord CG–299A/U.

3.4.2 *Frequency range.* The equipment shall be capable of measuring high frequency power over a frequency range extending from 2,400 mc to 3,333 mc.

3.4.3 *Sensitivity.* The equipment shall have a sensitivity of 2 milliwatts (mw) full scale on the indicating meter.

3.4.4 *Accuracy.* The accuracy of power measurement for an input power not exceeding 2 milliwatts shall be ±0.5 db over the temperature range of —4°F. to +125°F. and the frequency range of 2,400 mc to 3,333 megacycles.

3.4.5 *Antenna AT–67( )/AP VSWR.* The voltage standing wave ratio (VSWR) of Antenna AT–67( )/AP shall not be greater than 2.5 to 1 over the frequency range of 2,400 to 3,400 megacycles.

3.4.6 *Cable attenuation and calibration.* The nominal attenuation of the cable shall be held to 7 db ±0.5 db for a wavelength of 10 centimeters (cm). The accuracy of calibration of the cable shall be held to ±0.3 db of the nominal value.

3.5 Service conditions. The equipment shall meet the following service conditions:

3.5.1 *Equipment nonoperating.* Except as stated in 3.5.1(b), the equipment shall comply with the operational requirements of 3.5.2 after subjection to any of the following nonoperating conditions successively or in combination encountered during worldwide short-term storage and transit:

(a) *Temperature.* Exposure in the range of +160°F. to —80°F.; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time.

(b) *Relative humidity.* Relative humidity up to 97 percent, for an in-

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definite period of time; and exposure at 100-percent relative humidity with condensation, for 4 hours.

(c) *Elevation.* Elevation up to 25,000 feet above sea level.

(d) *Orientation.* Indefinite storage in any orientation.

(e) *Watertightness of case.* (see 4.8.4.1).

(f) *Rain.* Two inches of rainfall per hour, with wind at nominal 20 miles per hour.

(g) *Vibration, bounce, and shock.* As described in 3.6.

(h) *Salt spray.* See Specification MIL–T–945.

(i) *Fungus.* Subjected for extended periods to fungus and fungus supporting conditions.

3.5.2 *Equipment operating.* The equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) *Operation.* Operation for a minimum of 2,000 hours, with no more than normal maintenance and replacement of parts. Compliance will be determined by examination of the preproduction sample (see 3.8).

(b) *Temperature conditions, operating.* The equipment shall meet the operating requirements of this specification throughout the temperature range of —4°F. to +125°F.

(c) *Relative humidity.* Relative humidity up to 97 percent, and exposure at 100-percent relative humidity, with condensation, for 4 hours.

(d) *Elevation.* Elevation up to 10,000 feet above sea level.

(e) *Orientation.* Any orientation up to 20° from normal operating orientation.

(f) *Dust.* Dust as encountered during heavy traffic over dry clay roads.

3.6 Vibration, bounce, and shock.

3.6.1 *Vibration.* Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second when shock mounts (if any) are blocked or removed.

3.6.2 *Bounce and shock.* The equipment shall meet the requirements of table I.

TABLE I. *Bounce and shock*

Requirement	Insp. par.	Performance after inspection
Bounce test	4.8.1	Full specification performance. No physical damage except surface abrasion.
Shock; bench-handling	4.8.2	Full specification performance. No physical damage.
Shock; drop *	4.8.3	Equipment shall be operable. Minor mechanical damage is permissible.

* After this shock test, equipment performance shall conform to the following: Accuracy test (see 4.7.2.3.1).

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3.7 Preconditioning. Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.5 (also see 4.4(d)).

3.8 Preproduction samples. The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract (see 6.2(d)(1)).

3.9 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order (see 6.2).

3.10 Interchangeability. Corresponding components, replaceable subassemblies, and replaceable parts on contract shall be physically and functionally interchangeable as units without modification thereof or of other items with which the units are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.11 Workmanship. The equipment shall be manufactured and assembled in a thoroughly workmanlike manner, in accordance with the applicable portions of the following paragraphs:

(a) In Specification MIL–T–945:

General requirements for plastic material and plastic parts.

Wiring and cabling, including:

Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Flux and cleaning agents for soldering.
Process for soldering
Cleaning of equipment.
Riveting.
General requirements for of parts.
Cleaning of welding.
Process for welding.
Mechanical operation of controls.
Gears.
Self-tapping screws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery) (see 4.3).

(b) Acceptance inspection (acceptance inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.4 and 4.10.)

(1) Acceptance inspection of items before preparation for delivery (see 4.4).

(2) Acceptance inspection of preparation for delivery (see 4.10 and section 5).

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4.3 **Preproduction inspection.** This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table II and the group A, group B, and group C inspection specified in tables III, IV, and V, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling, (4) shock, drop, and (5) watertightness; other preproduction inspection may precede, follow or be interspersed between the foregoing.

TABLE II. *Preproduction inspection*

Inspection test	Rqt. par.	Test par.
Salt spray	3.5.1	4.8.9
Fungi	3.5.1	4.8.10
Temperature	3.5.2	4.8.5
Humidity	3.5.2	4.8.6
Vibration	3.6.1	4.8.8
Bounce	3.6.2	4.8.1
Shock, benchhandling	3.6.2	4.8.2
Shock, drop	3.6.2	4.8.3

4.4 **Acceptance inspection of items before preparation for delivery.** The minimum inspection, performed by the contractor shall be as specified by (a) through (d) below—to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government—at its discretion—will verify, by product inspection, the contractor's compliance with specified requirements (see 6.3). Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Acceptance inspection shall consist of group A, group B, and group C inspection as specified in 4.4.1 through 4.4.3.2.

(b) When Standard MIL–STD–105 specified actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied whtn it is considered more practical to select separate samples for group B or group C inspection, or both.

(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly (see 3.7).

4.4.1 *Group A inspection.* This inspection (including sampling) shall conform to table III and Standard MIL–STD–105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

4.4.2 *Group B inspection.* This inspection (including sampling) shall conform to table IV and the appendix to Standard MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.4.5 and the requirements of Standard MIL–STD–105 for disposition of rejected product.

4.4.2.1 *Group B sampling plans.* The group B sampling plans, as listed in table IV, shall be as follows:

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Group B plan	*AQL*	*Inspection level for normal inspection*	*Inspection level for reduced inspection*
B 1	4.0%	L–8	L–6
B 2	6.5%	L–7	L–5
B 3	10.0%	L–6	L–4
B 4	15.0%	L–5	L–3

TABLE III. *Group A inspection*

Inspection or test	Rqt. par.	Insp. par.	AQL	
			Major	Minor
Visual and mechanical	3.11	4.6		
Antenna AT–67( )/AP	3.2.4	4.6	1.0%	4.0%
Cord CG–299A/U(–ft –in)	3.2.3	4.6	1.0%	4.0%
Power meter TS–125( )/AP	3.2.1	4.6	1.5 dphu	6.5 dphu
Electrical:				
Antenna AT–67( )/AP, VSWR	3.4.5	4.7.3	1.0%	
Cord CG–299A/U (–ft –in).				
Attenuation and calibration	3.4.6	4.7.4	1.0%	
Frequency Range and Sensitivity	3.4.2 & 3.4.3	4.7.2.2	1.5 dphu	
Accuracy (77°F.)	3.4.4	4.7.2.3.1		
Power Range	3.4.1	4.7.2.4		
Antenna, immersion	3.2.4	4.8.4.2	1.0%	

4.4.2.2 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

4.4.2.3 *Procedure in case of failure.* When an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

TABLE IV. *Group B inspection*

Inspection or test	Reqt. par.	Insp. par.	Group B plan
Watertightness of battery housing cap	3.3.4	4.8.4.3	B–2
Watertightness of case for TS–125( )/AP	3.5.1	4.8.4.1	B–2
Interchangeability	3.10	4.9	B–2

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4.4.3 *Group C inspection.* This inspection shall be as listed in table V.

TABLE V. *Group C inspection*

Inspection or test	Reqt. par.	Insp. par.
Accuracy (—40°F. to +125°F.)	3.4.4	4.7.2.3.2

4.4.3.1 *Sampling plan.* One sample shall be selected for testing from the first production lot and then one sample from each 100 produced thereafter.

4.4.3.2 *Noncompliance.* If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.4 *Reinspection of conforming group B and group C sample units.* Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that thty are resubjected to and pass group A inspection after repair of all visible damage.

4.4.5 *Disposition of nonconforming product.* The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A or B inspection.

Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.5 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, N. Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.6 Visual and mechanical inspection. Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table VI.

TABLE VI. *Classification of visual and mechanical defects*

Major:

1. Cracks and/or holes in aluminum castings prior to application of finish. (Cracks over 1 inch long or $\frac{1}{32}$ inch wide, holes over $\frac{1}{16}$ inch in diameter.)
2. Hump in antenna plastic plate indicating too tight a fit or wave guide or rest for wave guide is too high.
3. Absence of nomenclature tag on Cord CG–299A/U (–ft. –in.).
4. Omission of attenuation and calibration on nomenclature tag for Cord CG–299A/U (–ft. –in.).
5. Balance control inoperative.
6. Balance control pin prohibiting closing of lid with control in on position inoperative.

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7. Stay chain missing or broken.
8. Improper fit or broken storage compartment caps.
9. Cracked meter glass.
10. MFP not applied to required points in an entire equipment.
11. Cracked or broken phenolic resistor board.
12. Missing components in lid or storage compartments.
13. Cover missing or inoperative.
14. Cracked or broken potentiometer housing or balance control knob.

Minor:

1. Cracks and/or holes in aluminum castings prior to application of finish when of insufficient magnitude to be considered major.
2. Balance control does not operate smoothly or freely.
3. Excess cement on antenna plastic plate and water seal gaskets.
4. Over or under sized water seal gaskets.
5. Bulging water seal gaskets due to excess tightening of hardware.
6. Uneasy operation of catches and handles.
7. Improper fit of components in case lid.
8. Stay chains not attached to caps or case.
9. Dust under glass of the meter.
10. MFP not applied at a particular required point.
11. Missing nameplate.

4.7 Electrical tests.

4.7.1 *Test equipment required.*

4.7.1.1 *Radio frequency signal generator.* The radio frequency signal generator shall consist of a tunable and stable oscillator with a square wave modulator, Sperry Gyroscope Company Klystron and Power Supply Signal Generator, or equal. The signal generator shall cover the frequency range of from 2,400 to 3,700 megacycles and shall have a power range of 30 to 250 milliwatts.

4.7.1.2 *Bolometer.* The bolometer, Sperry Gyroscope Company model 82X, or equal, and bolometer holder used in the test procedure shall be matched over the range of 2,400 to 3,700 megacycles.

4.7.1.3 *Power bridge.* The radio frequency power bridge shall be capable of indicating power levels to within at least 5 percent of its actual value when properly matched to the bolometer and holder.

4.7.1.4 *Attenuators.* The following attenuators shall be required:

(a) One attenuator variable over the range of 0 to 40 db and calibrated at 2,700 and 3,000 megacycles within ±0.2 db.

(b) The fixed attenuators calibrated to within ±0.2 db with a minimum value of at least 10 db.

4.7.1.4.1 *Calibration and marking.* Calibration of attenuators shall be marked on each attenuator. Calibration of attenuators for marking purposes shall be done at a center frequency of 2.7 kilamegacycles.

4.7.1.5 *Wavemeter.* Wavemeter Test Set TS–117( )/GP, or equal, calibrated to within ±3 megacycles of the actual frequency.

4.7.1.6 *Adapter.* Waveguide to coaxial type N fitting with a VSWR of 1.25 to 1 over the frequency range specified in 3.4.2.

4.7.1.7 *Coaxial switch.* General Communication Company, model 2N180MC, or equal.

4.7.1.8 *Probe and slotted section.* The probe and slotted section used in the test procedure shall be accurate to ±2 percent with bolometer output.

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4.7.1.9 *Standing wave amplifier.* Hewlett-Packard model 415A, or equal.

4.7.2 *Test procedures.*

4.7.2.1 *General.* The components of the test circuit shown in figure 1 shall be connected together with adapters; no cables shall be used in order to insure circuit stability. The value of attenuation in each of circuits 1 and 2 shall be determined to within 5 percent of the actual values. The attenuation of circuit 1 shall be measured from the coaxial switch terminal, through circuit 1, to the input terminal of Power Meter TS-125( )/AP; for circuit 2, it shall be measured from the coaxial switch terminal, through circuit 2, to the input terminal of the bolometer.

4.7.2.2 *Frequency range and sensitivity.*

(a) Adjust RF signal generator to 2,400 mc.

(b) Set coaxial switch to circuit 1 position.

(c) Adjust precision attenuator to indicate 2 milliwatts (full scale) on the Power Meter TS-125 ( )/AP.

(d) Set coaxial switch to circuit 2 position and tune bolometer for maximum power as indicated by a minimum VSWR reading.

(e) Record bolometer power value and compute actual power value considering value of calibrated attenuator.

(f) Repeat above test procedures for the frequency of 3,333 mc. The results shall conform to 3.4.2 and 3.4.3.

4.7.2.3 *Accuracy.*

4.7.2.3.1 *Accuracy test for group A inspection.* Test at room ambient temperature applying steps a through e of 4.7.2.2 at 2,400 mc and 3,333 mc and at any other frequency between 2,400 and 3,333 mc deemed necessary for compliance with 3.4.4.

4.7.2.3.2 *Accuracy test for group C inspection.* Test at room ambient temperature, —4° F. and +125°F. applying steps (a) through (e) of 4.7.2.2 at 2,400 mc and 3,333 mc and at any other frequency between 2,400 and 3,333 mc deemed necessary for compliance with 3.4.4.

4.7.2.4 *Power range.* The power range of the Power Meter TS-125( )/AP shall be verified by the measurement of the two attenuators, Attenuator CN-42( )/UP and Attenuator CN-43( )/UP, supplied as part of the Power Meter TS-125( )/AP. The test procedures shall be as follows:

(a) Set RF signal generator to a frequency between 2,400 and 3,333 mc.

(b) Set coaxial switch to circuit 2 position.

(c) Set precision attenuator so that the bolometer will read 10 milliwatts and record precision attenuator setting.

(d) Connect Attenuator CN-42( )/UP or Attenuator CN-43( )/UP between the output terminal of the coaxial switch and the input terminal of the calibrated attenuator in circuit 2.

(e) Set the precision attenuator so that bolometer will again read 10 milliwatts and record precision attenuator setting.

(f) From attenuator settings determine from the precision attenuator calibration chart, the value of attenuation.

(g) Repeat above procedure with second attenuator.

(h) Results shall conform to the requirements of 3.2.2.

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4.7.3 *VSWR of Antenna AT-67( )/AP.*

(a) Mount the antenna under test to the slotted section so that there are no reflecting objects in the path of its beam. Stand off to one side while performing the measurements.

(b) Set up the signal source for frequency of 2,400 mc in accordance with figure 2. Adjust the two output controls for maximum output.

(c) Turn the probe on the slotted section for maximum indication on the meter of the VSWR indicator using the 40 db scale. Adjust the probe for the minimum penetration into the slotted section required to give full scale reading at the 40 db setting.

(d) Move the probe carriage until a maximum reading is obtained. Adjust the gain control for a reading of unity on the VSWR scale.

Note. If the pointer cannot be brought to a unity indication, increase slightly the probe penetration, retune the probe readjust the gain control.

(e) Move the carriage until a minimum reading is obtained on the VSWR indicator. This reading represents the VSWR of the unit under test.

(f) Repeat procedure at 2,900 mc and 3,400 mc.

Freq. (mc)	VSWR (approx.)
2400	1.5
2900	2.0
3400	2.5

4.7.4 *Test for Cord CG-299A/U.* Cord CG-299A/U shall be tested and calibrated as follows:

(a) Set RF signal generator to a frequency between 2,400 mc and 3,333 mc.

(b) Set coaxial switch to circuit 2 position.

(c) Set precision attenuator so that the bolometer will read 10 milliwatts and record precision attenuator setting.

(d) Connect Cord CG-299A/U (-ft -in) between output terminal of coaxial switch and bolometer input terminal in circuit 2.

(e) Set the precision attenuator so that bolometer will again read 10 milliwatts and record precision attenuator setting.

(f) From the attenuator settings determine from the precision attenuator calibrator chart the value of attenuation.

(g) Results shall be 7 db ±0.3 db, in accordance with 3.4.6.

4.8 Service condition tests.

4.8.1 *Bounce test.* The equipment, on its shock mounts (if any) shall be secured to the Vehicular Adapter Plate and placed on the table of the Package Tester, both as made by the L.A.B. Corporation, Skaneateles, N. Y., or equal. The plate shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ±1 percent, for a total of 3 hours. The adapter plate shall be rotated through 90° at the end of each ¾-hour period, in the same direction.

4.8.2 *Shock test; bench-handling.* The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test

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shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30°, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "a", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.8.3 *Shock test; drop.* The equipment in its transit case shall be subjected to the drop test. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or a rigid steel frame. Height of drop shall be 48 inches. Drops shall be made on each face, edge, and corner for a total of 26 drops.

4.8.4 *Watertightness.*

4.8.4.1 Power Meter TS–125( )/AP shall be subjected to the watertight test specified in Standard MIL–STD–108. When the equipment case is opened there shall be no evidence of leakage.

4.8.4.2 *Antenna AT-67( )/AP.* Antenna AT–67/AP shall be immersed in 6 inches of water with 5 pounds of air pressure applied to the antenna connecting terminal for 1 minute, or until air bubbles cease to be given off by the water, whichever is longer. The immersed antenna shall be observed for evidence of leakage as indicated by bubbles emanating from the interior of the antenna. Bubbles which are the result of entrapped air on the various exterior parts shall not be considered as a leak. The antenna shall show no evidence of leakage.

4.8.4.3 *Battery housing cap.* The battery housing cap shall be subjected to the watertight test specified in Standard MIL–STD–108.

4.8.5 *Temperature test.* Power Meter TS–125( )/AP shall be subjected to the temperature range specified in 3.5.2(b) and tested as specified in Specification MIL–T–945.

4.8.6. *Relative humidity.* Power Meter TS–125( )/AP shall be subjected to the relative humidity specified in 3.5.2(c) and tested as specified in Specification MIL–T–945.

4.8.7 *Elevation.* Power Meter TS–125 ( )/AP shall be subjected to elevation conditions as specified in 3.5.2(d) and tested as specified in Specification MIL–T–945.

4.8.8 *Vibration.* Power Meter TS–125 ( )/AP shall be subjected to the vibration specified in 3.6.1 and test 2 of Specification MIL–T–945 and tested as specified in Specification MIL–T–945.

4.8.9 *Salt Spray.* Power Meter TS–125 ( )/AP shall be tested as specified in Specification MIL–T–945.

4.8.10 *Fungi test.* Power Meter TS–125 ( )/AP shall be tested as specified in Specification MIL–T–945.

4.9 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in

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cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate together, such as cases, covers, and mountings.

(d) Locations of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plugin assemblies and major components.

(e) Location of connectors, lockingpins, fasteners, slides, and mountings on plugin assemblies.

(f) Size and form of special threads.

4.10 Acceptance inspection of preparation for delivery. The minimum inspection performed by the contractor shall be as specified in 4.10.1 and 4.10.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government—at its discretion—will verify, by product inspection, the contractor's compliance with specified requirements (see 6.3). Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL–STD–105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.10.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in Specification MIL–P–116. Classification of defects shall be as shown in table VII.

4.10.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the appendix to Standard MIL–STD–105. Inspection level L–8 shall be used for normal inspection and L–6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R–1. The AQL for major defects shall be 4 percent and the AQL for minor defects shall be 10 percent. Classification of defects shall be as shown in table VIII.

4.10.2.1 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.10.2.2 *Procedure in case of failure.* If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.10.2.3 *Disposition of nonconforming product.* Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL–STD–105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.11 Rough handling test (preparation for delivery). When rough handling tests is required by the contract (see 6.2(d)), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling: Accuracy (see 4.7.2.3.1).

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TABLE VII. *Classification of defects for preservation, packaging and marking thereof*

Major:

1. Use of improper or defective material.
2. Quantity in unit package not as specified.
3. Incorrect packaging method applied.
4. Cushioning or padding omitted.
5. Cushioning inadequate for the physical and mechanical protection of the item.
6. Unsealed, puntured, or improperly sealed barrier bag, wrap, or envelope.
7. Stock number omitted, incorrect, or illegible.
8. Nomenclature omitted, incorrect, or illegible.
9. Marking of quantity of items in package omitted, incorrect, or illegible.
10. Different stock numbered items in the same unit package.

Minor:

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Packaging material damaged.
3. Air not expelled from barrier.
4. Insufficient area of barrier material provided to permit 2 subsequent seals.
5. Any item of marking information other than 7 through 9 omitted, incorrect, or illegible.

TABLE VIII. *Classification of defects for packing and marking thereof*

Major:

1. Use of improper or defective material.
2. Quantity in pack not as specified.
3. Gross weight in excess of specified amount.
4. Box closure not as specified.
5. Type, grade, class, and style of the shipping container not as specified.
6. Strapping omitted.
7. Strapping inadequate or incorrectly applied.
8. Items not adequately blocked, braced, or cushioned with the shipping container to prevent movement or damage.
9. Shipping documents or packing list omitted.
10. Stock number omitted, incorrect, or illegible.
11. Nomenclature omitted, incorrect, or illegible.
12. Marking of quantity of items in pack omitted, incorrect, or illegible.
13. Destination marking omitted, incorrect, or illegible.
14. Service designation (color marking) omitted.
15. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
16. Overseas code marking omitted, incorrect, or illegible.

Minor:

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL–STD–105.
4. Any item of required marking information other than 10 through 16 listed under major defects omitted, incorrect, or illegible.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Power Meter TS–125 ( )/AP shall be preserved, packaged, and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116 and described herein.

5.1.1.1 *Technical manual.* Each technical manual shall be packaged method IC–3, as follows: Inclose each technical manual within a close-fitting, type II, class "b" bag, conforming to Specification MIL–B–117. Heat seal bag to effect closure.

5.1.1.2 *Antenna, horn, attenuators, resistor unit, cable assembly.* No packaging required. Stow and secure the listed items within the case cover of the power meter.

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Fill all voids with type 1, wrapping cushioning, paperboard, conforming to Specification PPP-B-291. Position the cover and secure all fastenings.

5.1.1.3 *Unit package.*

5.1.1.3.1 *Power Meter TS-125( )/AP.* Each power meter shall be unit packaged method IC-5 as follows: Cushion each power meter on all surfaces with cells or pads or both, fabricated of fiberboard, having a minimum dry bursting strength of 175 pounds, and designed to absorb the shock of impact encountered in handling and transit. Cells and pads should have a water resistance equal to that of the box in which they are used. Place each cushioned power meter within a close-fitting, regular slotted style, type I, class 2, grade 5, box, conforming to Specification PPP-B-636. Seal the box as specified in the appendix of the box specification. Place the packaged technical manual on the lid of the box and secure in position with waterproof, pressure-sensitive tape, type III, class 3, conforming to Specification PPP-T-60.

5.1.2 *Level C.* Power meter TS-125( )/AP shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration, and physical damage, during direct shipment to the first receiving activity.

5.2 **Packing.**

5.2.1 *Level A.* The Power Meter packaged as specified (see 6.2) shall be packed within class 2, style 4, nailed, wood boxes, conforming to Specification PPP-B-621, or wood wirebound boxes, class 3, conforming to Specification PPP-B-585, or boxes, wood-cleated fiberboard overseas type, conforming to Specification PPP-B-591, or wood, cleated plywood boxes conforming to Specification PPP-B-601 style A or B; the plywood shall be container grade conforming to Specification NN-P-515 type II or III, class 2, or fiber boxes conforming to Specification PPP-B-63C class 2. Fabricate the boxes to fit the contents snugly. Containers shall be closed in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped with flat steel strapping, in conformance with the requirements of the container specification.

5.2.2 *Level B.* The Power Meter, packaged as specified (see 6.2) shall be packed as specified in 5.2.1 except that the box shall conform to class 1, style 4, of Specification PPP-B-621, or wood wirebound boxes, class 1, conforming to Specification PPP-B-585, or boxes, wood-cleated fiberboard conforming to Specification PPP-B-591 (domestic type), or wood, cleated plywood boxes conforming to Specification PPP-B-601 (domestic type), or fiber boxes conforming to Specification PPP-B-636 class 1. Closure requirements shall be in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed approximately 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* The power meter, packaged as specified (see 6.2) shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

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5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Power Meter TS-125( )/AP measures cw, modulated or pulsed average radio frequency (RF) power of radar transmitters. Measurements from 0 to 2 milliwatts with direct coupling to transmitter, up to 4 watts by using attenuators, and above 4 watts by using pickup antenna.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (level A, level B, or level C).

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

(d) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample.

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon—including rough handling test, which will not be performed as procurement inspection.

(d) Five copies of process sheet, using format shown on figure 3 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.

(e) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required.

(h) Quantity of tools and running spare parts required (see 3.9).

(i) Submission of the statement of treatment as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Verification inspection. The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) *Type A.* The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Standard MIL-STD-105.

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(b) *Type B.* That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Standard MIL–STD–105.

(c) *Type C.* A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: For example Power Meter TS–125W/AP. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information (see 1.1).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Navy—Bureau of Aeronautics
Air Force—WADC

Preparing activity:
Army—Signal Corps

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Power Meter TS-125( )/AP

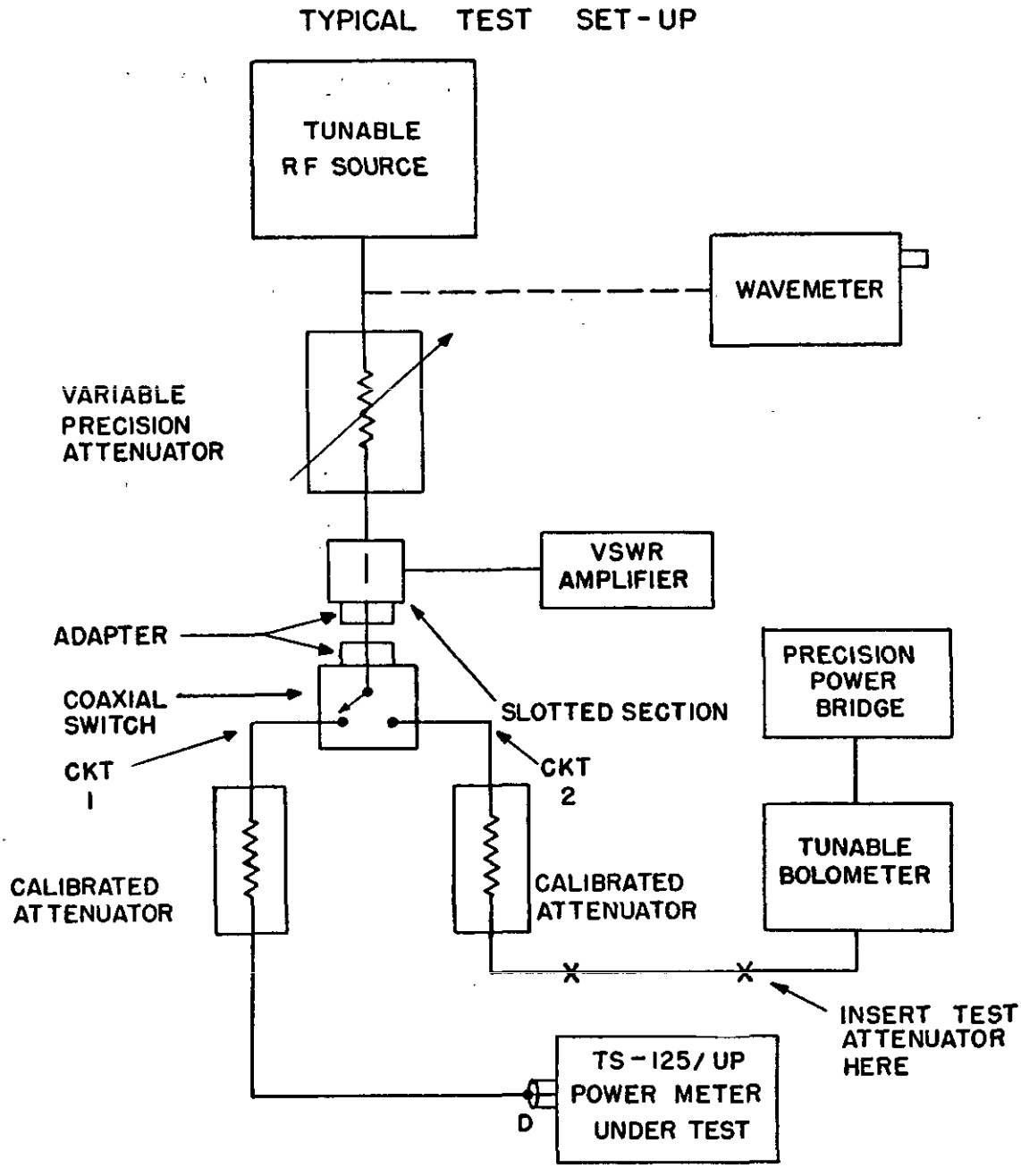


FIGURE 1.

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Spec No.
Mil-P-11592B Power Meter TS-125()/AP

Antenna VSWR Test Set-up.

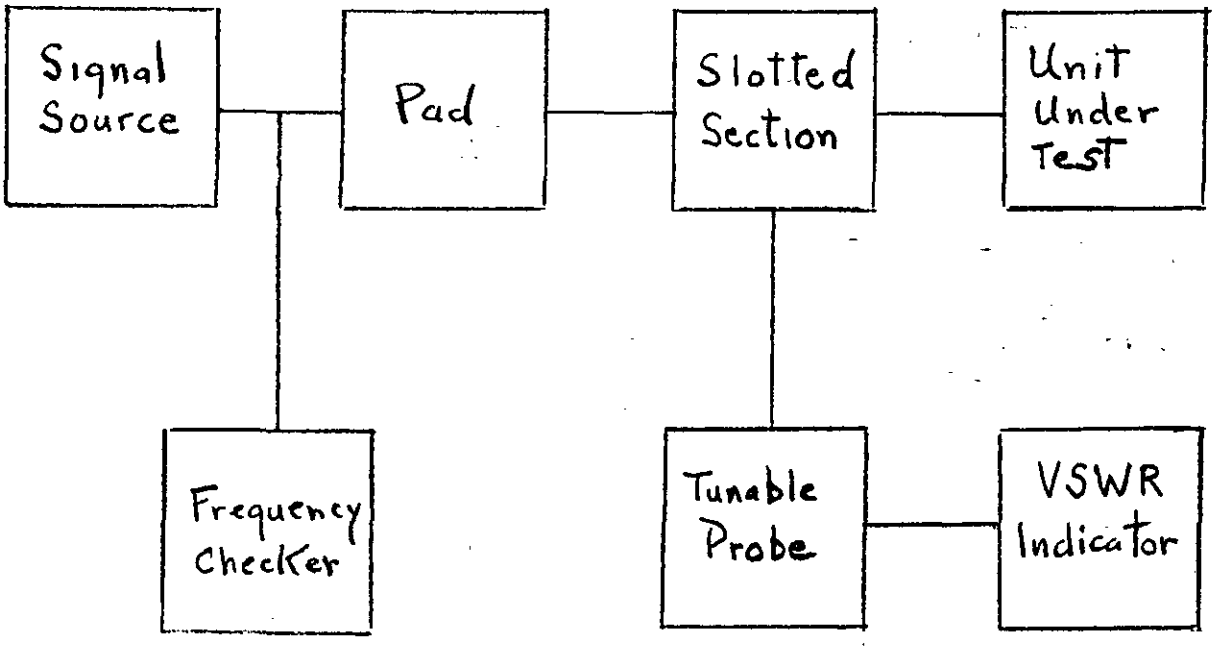


FIGURE 2.

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PACKAGING, PACKING, AND MARKING PILOT PACK PROCESS SHEET

CONTRACTOR
STREET ADDRESS
CITY AND STATE

ORDER NO._____
EQUIPMENT_____
LEVEL_____

DATE PILOT PACK COMPLETED FOR EVALUATION_____

ITEM NAME											
PACKAGING METHOD											
CLEANING	SPECIFICATION MATERIAL DESCRIPTION	T		G		C		T		G	
PRESERVATION	SPECIFICATION MATERIAL DESCRIPTION	T		G		C		T		G	
CUSHIONING	SPECIFICATION MATERIAL DESCRIPTION SIZE	T		G		C		T		G	
SECURING	SPECIFICATION MATERIAL DESCRIPTION SIZE	T		G		C		T		G	
BARRIER	SPECIFICATION MATERIAL DESCRIPTION SIZE	T		G		C		T		G	
DESICCANT	SPECIFICATION MATERIAL DESCRIPTION SIZE	T		G		C		T		G	
UNIT CARTON	SPECIFICATION MATERIAL DESCRIPTION SIZE	T		G		C		T		G	
INTERMEDIATE CARTON	SPECIFICATION MATERIAL DESCRIPTION SIZE	T		G		C		T		G	
OUTER CARTON (CONSOLIDATED OR 11b)	SPECIFICATION MATERIAL DESCRIPTION SIZE	T		G		C		T		G	
SECURE CLOSURE	SPECIFICATION MATERIAL DESCRIPTION SIZE	T		G		C		T		G	

CASE LINER	SPECIFICATION SEALING MATERIAL SIZE	
SHIPPING CONTAINER	SPECIFICATION STYLE SIZE	
STRAPPING	SPECIFICATION MATERIAL DESCRIPTION SIZE	

LEGEND FOR MATERIAL: T-TYPE, G-GRADE, C-CLASS

FIGURE 3.

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